

# QUESTION BANK

**Course: EEUC201 — Analog & Digital Electronics**

Semester 3rd | Academic Year 2026-27

## Course Outcomes (COs)

**CO1:** Design & Enhance their skills in basics of analog and digital circuits.

**CO2:** Evaluate the performance and classification of various components and ICs.

**CO3:** State linear and non-linear applications of operational amplifiers.

**CO4:** Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates (combinational circuits), timer and counters (sequential circuits).

**CO5:** Analyze characteristics of popular MOS and bipolar logic families, with emphasis on CMOS and TTL technologies.

## Mapping of Course Outcomes with Program Outcomes

| CO  | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 3   | 2   | -   | -   | -   | -   | -   | -   | -   | -    | -    | -    | 3    | 1    |
| CO2 | 3   | 3   | -   | -   | -   | -   | -   | -   | -   | -    | -    | -    | 2    | -    |
| CO3 | 2   | 3   | -   | -   | -   | -   | -   | -   | -   | -    | -    | -    | 2    | -    |
| CO4 | 1   | 2   | 2   | 3   | 3   | -   | -   | -   | -   | -    | -    | -    | 3    | 1    |
| CO5 | 1   | 1   | 3   | 3   | -   | -   | -   | -   | -   | -    | -    | -    | 2    | 1    |

*Note: Each question in the bank below is tagged with the Unit it belongs to, the Course Outcome(s) (CO) it primarily addresses, the corresponding Program/Program-Specific Outcomes (PO/PSO) attained through that CO (with mapping strength in parentheses, per the table above), and an indicative Bloom's Taxonomy level. Unit-to-CO correspondence follows the course syllabus and instructional design; instructors may refine individual mappings as needed.*

## Question Bank

| Sr. No.                    | Question                                                                                                                                  | CO Mapped | PO / PSO Mapped                  | Bloom's Level |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------|---------------|
| <b>Unit 1: BJT and FET</b> |                                                                                                                                           |           |                                  |               |
| 1                          | Which transistor connection is commonly used and why?                                                                                     | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| 2                          | Define current amplification factor $\alpha$ , $\beta$ , and $\gamma$ . Also give their relationship in terms of mathematical expression. | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| 3                          | What is operating point?                                                                                                                  | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Remember      |
| 4                          | What is a d.c load line? What is its importance?                                                                                          | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Remember      |
| 5                          | What is faithful amplification? How can it be achieved in a transistor amplifier?                                                         | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Remember      |
| 6                          | What is transistor biasing?                                                                                                               | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Remember      |
| 7                          | What are the methods of transistor biasing?                                                                                               | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Remember      |
| 8                          | Explain the base resistor method of biasing?                                                                                              | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| 9                          | What do you mean by stabilization of operating point? What is its need?                                                                   | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Remember      |
| 10                         | What is stability factor? Derive the equation of stability factor                                                                         | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Apply         |
| 11                         | Describe the feedback resistor method of biasing                                                                                          | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| 12                         | Describe the potential divider method of biasing.                                                                                         | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| 13                         | Describe the steps to be taken in the design of biasing circuits for low powered transistors.                                             | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| 14                         | Examples of Bipolar Junction Transistor                                                                                                   | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| 15                         | What is the origin of the name FET?                                                                                                       | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Remember      |
| 16                         | Explain the construction and working of JFET                                                                                              | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| 17                         | Why FET is called a unipolar transistor?                                                                                                  | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| 18                         | Give the difference between JFET and BJT.                                                                                                 | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Analyze       |
| 19                         | How will you determine the output characteristics of a JFET?                                                                              | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| 20                         | Explain the terms (i) shorted gate drain current (ii) Pinch off voltage (iii) gate-source cut off voltage as applied to FET.              | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| 21                         | Give the expression for drain current in terms of $I_{DSS}$ , $V_{GS}$ and $V_{GS(off)}$ .                                                | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| 22                         | What is MOSFET? Give its constructional details.                                                                                          | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| 23                         | Describe the working of MOSFET.                                                                                                           | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |

| Sr. No.                               | Question                                                                                                                                                                                                                                          | CO Mapped | PO / PSO Mapped                  | Bloom's Level |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------|---------------|
| 24                                    | Write the application of MOSFET                                                                                                                                                                                                                   | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| 25                                    | Why is the input impedance of a MOSFET higher than that of a FET?                                                                                                                                                                                 | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| <b>Unit 2: Operational Amplifiers</b> |                                                                                                                                                                                                                                                   |           |                                  |               |
| 26                                    | What is an op-amp? List the four basic building blocks of an op-amp                                                                                                                                                                               | CO2       | PO1(3), PO2(3), PSO1(2)          | Understand    |
| 27                                    | State the applications of OP-AMP.                                                                                                                                                                                                                 | CO2       | PO1(3), PO2(3), PSO1(2)          | Understand    |
| 28                                    | Define and explain ideal differential amplifier.                                                                                                                                                                                                  | CO2       | PO1(3), PO2(3), PSO1(2)          | Understand    |
| 29                                    | Draw the block diagram of an OP-AMP and explain the purpose of using each block.                                                                                                                                                                  | CO2       | PO1(3), PO2(3), PSO1(2)          | Understand    |
| 30                                    | Write Features of differential amplifier.                                                                                                                                                                                                         | CO2       | PO1(3), PO2(3), PSO1(2)          | Understand    |
| 31                                    | Draw neat circuit diagram of dual input balanced output differential amplifier.                                                                                                                                                                   | CO2       | PO1(3), PO2(3), PSO1(2)          | Understand    |
| 32                                    | Draw neat circuit diagram of dual input un-balanced output differential amplifier.                                                                                                                                                                | CO2       | PO1(3), PO2(3), PSO1(2)          | Understand    |
| 33                                    | Draw pin diagram of IC 741.                                                                                                                                                                                                                       | CO2       | PO1(3), PO2(3), PSO1(2)          | Understand    |
| 34                                    | Define the parameter of an Op-amp. (i) Input resistance (ii) Input offset voltage Common mode rejection ratio(CMRR) (iv) Power supply rejection ratio(PSRR) (v) Slew rate(SR) (vi) Input bias current (vii) Input offset current (viii) Bandwidth | CO2       | PO1(3), PO2(3), PSO1(2)          | Remember      |
| 35                                    | What are the characteristics of an ideal op-amp?                                                                                                                                                                                                  | CO2       | PO1(3), PO2(3), PSO1(2)          | Remember      |
| <b>Unit 3: Applications of Op-amp</b> |                                                                                                                                                                                                                                                   |           |                                  |               |
| 36                                    | Explain why op-amp is not used as amplifier in the open loop mode. Or Give two reasons why an open-loop op-amp is unsuitable for linear applications.                                                                                             | CO3       | PO1(2), PO2(3), PSO1(2)          | Understand    |
| 37                                    | What is the difference between positive and negative feedback?                                                                                                                                                                                    | CO3       | PO1(2), PO2(3), PSO1(2)          | Analyze       |
| 38                                    | Explain why is negative feedback preferred in op-amp used as amplifier?                                                                                                                                                                           | CO3       | PO1(2), PO2(3), PSO1(2)          | Understand    |
| 39                                    | What are the advantages of using feedback?                                                                                                                                                                                                        | CO3       | PO1(2), PO2(3), PSO1(2)          | Remember      |
| 40                                    | State the application of OP-AMP.                                                                                                                                                                                                                  | CO3       | PO1(2), PO2(3), PSO1(2)          | Understand    |
| 41                                    | Explain with the help of circuit diagram the operation of a closed-loop op-amp, non-inverting amplifier with negative feedback. Derive an expression for the voltage gain of this amplifier. (J.S.Katre)                                          | CO3       | PO1(2), PO2(3), PSO1(2)          | Apply         |
| 42                                    | Explain with the help of circuit diagram the operation of an op-amp, inverting amplifier with negative feedback. Derive an expression for the voltage gain of this amplifier. (J.S.Katre)                                                         | CO3       | PO1(2), PO2(3), PSO1(2)          | Apply         |
| 43                                    | Draw and explain the voltage follower using OP-AMP.                                                                                                                                                                                               | CO3       | PO1(2), PO2(3), PSO1(2)          | Understand    |
| 44                                    | Draw a neat circuit diagram of op-amp as an Inverter?                                                                                                                                                                                             | CO3       | PO1(2), PO2(3), PSO1(2)          | Understand    |

| Sr. No. | Question                                                                                                                                | CO Mapped | PO / PSO Mapped                                                                    | Bloom's Level |
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| 45      | Draw and explain differential amplifier with one OP-AMP.                                                                                | CO3       | PO1(2), PO2(3), PSO1(2)                                                            | Understand    |
| 46      | Explain OP-AMP is used as a summing amplifier                                                                                           | CO3       | PO1(2), PO2(3), PSO1(2)                                                            | Understand    |
| 47      | Explain OP-AMP is used as a scaling amplifier                                                                                           | CO3       | PO1(2), PO2(3), PSO1(2)                                                            | Understand    |
| 48      | Explain OP-AMP is used as a averaging amplifier                                                                                         | CO3       | PO1(2), PO2(3), PSO1(2)                                                            | Understand    |
| 49      | Give the comparison of inverting and non-inverting amplifiers in closed loop mode.                                                      | CO3       | PO1(2), PO2(3), PSO1(2)                                                            | Analyze       |
| 50      | What is an instrumentation amplifier? List three applications of the instrumentation amplifier.                                         | CO3       | PO1(2), PO2(3), PSO1(2)                                                            | Understand    |
| 51      | Draw and explain the circuit of current to voltage (I to V) converter. Mention the field of application.                                | CO3       | PO1(2), PO2(3), PSO1(2)                                                            | Understand    |
| 52      | Explain the circuit of voltage to current converter (V to I converter) with floating load. Mention the field of application.(j.s.Katre) | CO3       | PO1(2), PO2(3), PSO1(2)                                                            | Understand    |
| 53      | What are the types of comparator? Explain the operation of an inverting comparator. Draw input and output voltage waveforms.            | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Understand    |
| 54      | What are the types of comparator? Explain the operation of a Non-inverting comparator. Draw input and output voltage waveforms.         | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Understand    |
| 55      | Explain the operation of a zero crossing detector.                                                                                      | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Understand    |
| 56      | Draw and explain the circuit of differentiator using OP-AMP                                                                             | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Understand    |
| 57      | Draw and explain the circuit of Integrator using OP-AMP                                                                                 | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Understand    |
| 58      | State the applications of comparator and explain any one of them.                                                                       | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Understand    |
| 59      | Draw the circuit of a Schmitt trigger and explain its working.                                                                          | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Understand    |

| Sr. No. | Question                                                                                                                                                                                                                                  | CO Mapped | PO / PSO Mapped                                                                    | Bloom's Level |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------|---------------|
| 60      | What is the basic difference between comparator and Schmitt trigger.                                                                                                                                                                      | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Analyze       |
| 99      | What do you mean by active filter? What are its advantages over passive filter? What are its disadvantages?                                                                                                                               | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Remember      |
| 100     | Define filter and give classification of filters                                                                                                                                                                                          | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Understand    |
| 101     | Compare active and passive filters.                                                                                                                                                                                                       | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Analyze       |
| 102     | Explain with the help of circuit diagram, the operation of first order low-pass filter.                                                                                                                                                   | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Understand    |
| 103     | Explain with the help of circuit diagram, the operation of first order High-pass filter.                                                                                                                                                  | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Understand    |
| 104     | Explain with the help of circuit diagram, the operation of Second order low-pass filter.                                                                                                                                                  | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Understand    |
| 105     | Explain with the help of circuit diagram, the operation of Second order High-pass filter.                                                                                                                                                 | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Understand    |
| 106     | Design a first order low-pass filter for the following specifications: Pass-band voltage gain =2 Cut-off frequency $f_c = 10\text{KHz}$ , Choose capacitance $C = 0.001\ \mu\text{F}$ .                                                   | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Create/Apply  |
| 107     | For a first order high pass filter, calculate the value of R if $C = 0.0047\ \mu\text{F}$ and $f_c = 10\text{KHz}$ .                                                                                                                      | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Apply         |
| 108     | For a first order high pass filter, calculate cut-off frequencies $f_c$ if the component values are $R = 15\text{K}\Omega$ and $C = 0.01\ \mu\text{F}$ . calculate the pass-band gain if $R_f = 10\text{Kohm}$ and $R_1 = 5\text{Kohm}$ . | CO4, CO3  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(2), PO2(3), PSO1(2) | Apply         |
| 109     | Explain circuit of positive clippers with input and output waveform.                                                                                                                                                                      | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)                           | Understand    |

| Sr. No.                                                    | Question                                                             | CO Mapped | PO / PSO Mapped                                                                     | Bloom's Level |
|------------------------------------------------------------|----------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------|---------------|
| 110                                                        | Explain circuit of negative clippers with input and output waveform. | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)                            | Understand    |
| 111                                                        | Explain positive and negative clampers.                              | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)                            | Understand    |
| <b>Unit 4: Specialized ICs</b>                             |                                                                      |           |                                                                                     |               |
| 61                                                         | Explain 555 timer IC as a monostable multivibrator.                  | CO4, CO2  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(3), PO2(3), PSO1(2)  | Understand    |
| 62                                                         | Explain 555 timer IC as a astable multivibrator                      | CO4, CO2  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(3), PO2(3), PSO1(2)  | Understand    |
| 63                                                         | Explain 555 timer IC as a bistable multivibrator                     | CO4, CO2  | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1)   PO1(3), PO2(3), PSO1(2)  | Understand    |
| 64                                                         | Give the different types of voltage regulator with their IC Number.  | CO2       | PO1(3), PO2(3), PSO1(2)                                                             | Understand    |
| <b>Unit 5: Digital Electronics &amp; Numbering Systems</b> |                                                                      |           |                                                                                     |               |
| 65                                                         | Examples of Number system                                            | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1)                                                    | Understand    |
| 66                                                         | Examples of Binary arithmetic                                        | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1)                                                    | Understand    |
| 67                                                         | Examples of 1's complement                                           | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1)                                                    | Understand    |
| 68                                                         | Examples of 2's complement                                           | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1)                                                    | Understand    |
| 69                                                         | BCD code conversion                                                  | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1)                                                    | Understand    |
| 70                                                         | Grey code conversion & its application                               | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1)                                                    | Understand    |
| <b>Unit 6: Digital Logic Families and Design</b>           |                                                                      |           |                                                                                     |               |
| 71                                                         | Explain laws of Boolean algebra.                                     | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1)                                                    | Understand    |
| 72                                                         | Explain De- Morgan's Theorem.                                        | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1)                                                    | Understand    |
| 73                                                         | Why are NAND and NOR gates called universal gates?                   | CO5, CO1  | PO1(1), PO2(1), PO3(3), PO4(3), PSO1(2), PSO2(1)   PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |
| 74                                                         | Show that both NAND gate and NOR gate are universal gates.           | CO5, CO1  | PO1(1), PO2(1), PO3(3), PO4(3), PSO1(2), PSO2(1)   PO1(3), PO2(2), PSO1(3), PSO2(1) | Understand    |

| Sr. No.                            | Question                                                                                                          | CO Mapped | PO / PSO Mapped                                          | Bloom's Level |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------|---------------|
| 75                                 | Realize X-OR operation using (a) Only NAND gates (b) Only NOR gates.                                              | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1)                         | Understand    |
| 76                                 | Examples of Reduce the Boolean expressions & Construct logic circuit diagram according to Boolean expression.     | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1)                         | Understand    |
| 77                                 | Karnaugh Map (Kmap) SOP form POS form                                                                             | CO1       | PO1(3), PO2(2), PSO1(3), PSO2(1)                         | Understand    |
| 78                                 | Explain Diode Transistor Logic.                                                                                   | CO5       | PO1(1), PO2(1), PO3(3), PO4(3), PSO1(2), PSO2(1)         | Understand    |
| 79                                 | Explain Resistor Transistor Logic.                                                                                | CO5       | PO1(1), PO2(1), PO3(3), PO4(3), PSO1(2), PSO2(1)         | Understand    |
| 80                                 | Explain integrated injection Logic.                                                                               | CO5       | PO1(1), PO2(1), PO3(3), PO4(3), PSO1(2), PSO2(1)         | Understand    |
| 81                                 | With the help of neat circuit diagram, explain the working of (a) a TTL NAND logic(b) TTL NOR logic               | CO5       | PO1(1), PO2(1), PO3(3), PO4(3), PSO1(2), PSO2(1)         | Understand    |
| 82                                 | With the help of neat circuit diagram, explain the working of (a) a CMOS inverter (b) a two-input CMOS NAND gate. | CO5       | PO1(1), PO2(1), PO3(3), PO4(3), PSO1(2), PSO2(1)         | Understand    |
| 83                                 | Define the terms: Propagation delay Fan-in Fan-out Power dissipation Noise margin                                 | CO5       | PO1(1), PO2(1), PO3(3), PO4(3), PSO1(2), PSO2(1)         | Remember      |
| <b>Unit 7: Combinational Logic</b> |                                                                                                                   |           |                                                          |               |
| 84                                 | Explain the working of a half adder? Draw its logic diagram.                                                      | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1) | Understand    |
| 85                                 | Explain the working of a Full adder? Draw its logic diagram.                                                      | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1) | Understand    |
| 86                                 | What is half sub tractor? Draw the logic diagram and truth table of half sub tractor.                             | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1) | Remember      |
| 87                                 | Explain and design full subtractor circuit.                                                                       | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1) | Understand    |
| 88                                 | What is meant by a multiplexer? Explain with block diagram the principle of multiplexer.                          | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1) | Understand    |
| 89                                 | What is meant by a demultiplexer? Explain with block diagram the principle of demultiplexer.                      | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1) | Understand    |
| 90                                 | Distinguish between multiplexer and de-multiplexer.                                                               | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1) | Analyze       |

| Sr. No.                                        | Question                                                                                                                       | CO Mapped | PO / PSO Mapped                                          | Bloom's Level |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------|---------------|
| 91                                             | What do you mean by a priority encoder? Give example.                                                                          | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1) | Understand    |
| 92                                             | Explain BCD to 7 segment decoders.                                                                                             | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1) | Understand    |
| <b>Unit 8: Flip-flops, Timers and Counters</b> |                                                                                                                                |           |                                                          |               |
| 93                                             | Draw the circuit diagram of a positive level triggered S-R Flip Flop and explain its operation with the help of a truth table. | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1) | Understand    |
| 94                                             | Write notes on edge-triggered J-K flip-flop .                                                                                  | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1) | Understand    |
| 95                                             | Write notes on Level Triggered D Flip-Flop.                                                                                    | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1) | Understand    |
| 96                                             | Write notes on level Triggered J-K Flip-flop.                                                                                  | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1) | Understand    |
| 97                                             | Give the comparison between Asynchronous and Synchronous counters                                                              | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1) | Understand    |
| 98                                             | Give the comparison between Registers and Counters.                                                                            | CO4       | PO1(1), PO2(2), PO3(2), PO4(3), PO5(3), PSO1(3), PSO2(1) | Understand    |